

KEERTHANA BELLAM

Generative AI Engineer | ML Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)



PROFESSIONAL SUMMARY

Generative AI and ML Engineer with 3+ years of experience architecting and deploying LLM-powered applications, Retrieval-Augmented Generation (RAG) pipelines, and semantic search systems across 500+ documents and 100K+ customer records. Proficient in Python, LangChain, OpenAI and Google Gemini APIs, FAISS, prompt engineering, and multi-agent workflows (LangGraph, CrewAI). Skilled in building and deploying end-to-end, production-ready AI applications with FastAPI, Streamlit, Docker, and Kubernetes, translating complex business requirements into scalable AI solutions that cut manual review and reporting time from hours to minutes.

EXPERIENCE

AI Analyst

August 2024 – Present

Workada, USA

- Analyzed internal business workflows across 2 core workflows, identifying AI-driven automation opportunities for reporting, analysis, and recurring operational processes, cutting manual processing time by roughly 1 hour per day.
- Developed AI/ML-assisted reporting workflows across 8+ recurring business reports, shortening reporting turnaround from about 3 business days to under 2 business days.
- Designed prompt-engineering and model-evaluation workflows with structured evaluation criteria across 15+ test scenarios, reducing manual review time from roughly 4 hours to 3 hours per cycle.
- Built documentation and standardized procedures for AI-assisted workflows, shortening new-hire onboarding from 4 weeks to 3 weeks.
- Monitored AI-generated outputs across 5 production workflows, identifying and correcting an average of 8 recurring model errors per week through prompt refinement, validation, and workflow improvements.

Office & Data Automation Assistant

January 2025 – May 2026

Rowan University Campus Recreation Center

- Developed data automation workflows across 4 operational data areas — membership, attendance, transactions, and reporting datasets spanning 1,000+ records — standardizing data collection, validation, reconciliation, and recurring reporting.
- Built automated Excel KPI dashboards using Pivot Tables, VLOOKUP, formulas, and trend analysis, consolidating 6 manual reports into 2 automated dashboards for recurring operational reporting.
- Automated repetitive data-preparation and reporting processes using Excel automation and AI-assisted tools, cutting weekly reporting effort from 6 hours to about 4 hours.

ML Engineer

June 2022 – July 2024

BuiltInTech, Hyderabad, India

- Developed an end-to-end customer churn prediction pipeline using Python, Pandas, NumPy, and SQL across 100K+ customer accounts, covering preprocessing, feature engineering, model training, validation, and deployment.
- Engineered behavioral and transactional features and trained 3 classification models — Random Forest, Decision Tree, and XGBoost — reaching an accuracy score of 0.88 and an AUC of 0.89 for customer churn prediction.
- Performed hyperparameter tuning using grid/random search and cross-validation across 3 ML models, raising F1 score from 0.79 to 0.87, an 8-point gain in churn classification performance.

PROJECTS

Ask-Keerthana-AI

PYTHON • STREAMLIT • OPENAI API

- Built a conversational AI portfolio assistant using Python, Streamlit, and OpenAI API, providing grounded responses across education, professional experience, technical skills, and project information.
- Designed structured prompting and response-grounding logic to keep answers aligned with the portfolio knowledge base, with a contact fallback for roughly 1 in 20 unanswerable questions.
- Evaluated the assistant using 50+ representative user questions, achieving on-topic, accurate responses for 47 of 50 questions and reducing off-topic responses from 15 to 5.

GenAI Resume Analyzer

RAG • LANGCHAIN • OPENAI • FAISS • STREAMLIT

- Architected an end-to-end RAG-based resume analysis platform using LangChain, OpenAI, FAISS, embeddings, and Streamlit to process 500+ resumes and job descriptions through document parsing, chunking, semantic retrieval, and LLM generation.
- Built a Streamlit application supporting document upload, JD analysis, semantic retrieval, ATS scoring, skill-gap analysis, and AI-generated recommendations, cutting manual screening time per resume from 10 minutes to 5 minutes.

GenAI-Powered Data Analyst Assistant

PYTHON • PANDAS • STREAMLIT • GOOGLE GEMINI API

- Built an AI-powered data-analysis application using Python, Pandas, NumPy, Streamlit, and Google Gemini API to automate profiling, exploratory analysis, and statistical summaries across 200+ datasets.
- Built an interactive Streamlit workflow supporting data upload, automated EDA, visualizations, AI-powered Q&A, and downloadable reports, cutting manual EDA effort from about 2 hours to under 1 hour.

SKILLS

Programming	Python, R, SQL, JavaScript, C++, Git, GitHub
Generative AI & LLM Engineering	Generative AI, LLMs, NLP, Prompt Engineering, LLM Integration, Function Calling, AI Agents, Agentic Workflows (LangGraph, CrewAI), Guardrails, Responsible AI
RAG & Retrieval	RAG, Embeddings, Vector Search, Semantic Search, Reranking, FAISS, Pinecone, ChromaDB, Milvus, Weaviate, Neo4j, Elasticsearch
Machine Learning & Deep Learning	Classification, Regression, Clustering, Random Forest, XGBoost, LightGBM, Scikit-learn, Feature Engineering, Hyperparameter Tuning, Time Series Forecasting, Anomaly Detection, Statistical Modeling, Deep Learning, CNN, Transformer Models, PyTorch, TensorFlow, Hugging Face Transformers, BERT, Model Evaluation
Model Evaluation & Governance	A/B Testing, Hypothesis Testing, SHAP, LIME, ROC-AUC, Precision/Recall, RMSE, MAPE
Data Engineering & Visualization	ETL, Data Pipelines, Apache Spark, Kafka, Airflow, Delta Lake, Pandas, NumPy, Exploratory Data Analysis, Tableau, Power BI
Backend & Cloud	FastAPI, Flask, REST APIs, Docker, Kubernetes, CI/CD, Jenkins, MLflow, Model Monitoring, Experiment Tracking, AWS (S3, EC2, Lambda, SageMaker), Amazon Bedrock, Azure, Azure ML, GCP Vertex AI, BigQuery
Databases & Collaboration	MySQL, PostgreSQL, Cassandra, Snowflake, Astra DB, Redis, Jira, Confluence, Notion, Agile, SDLC

CERTIFICATIONS

- AWS Certified Generative AI Developer – Professional
- AWS Certified AI Practitioner
- IBM SkillsBuild — Getting Started with Generative AI

EDUCATION

Rowan University — Master of Science in Data Science
Koneru Lakshmaiah Educational Foundation — Bachelor of Technology in Computer Science Engineering